Lecture Notes for

Neural Networks I

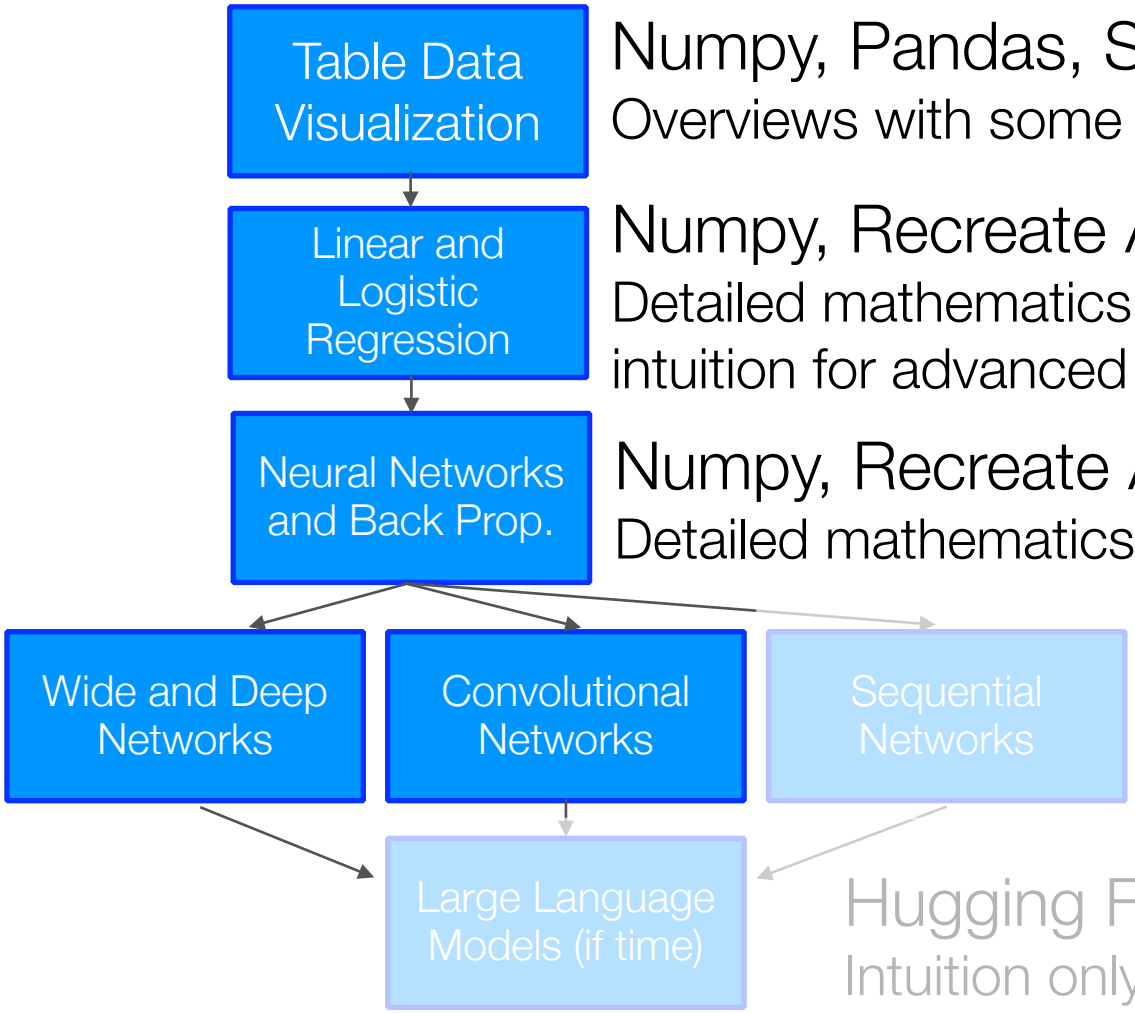


Convolutional Neural Networks

Logistics and Agenda

- Logistics
  - Deep Learning Lab due soon!
  - Town Hall, if needed
- Agenda
  - Basic CNN Structures
  - CNN Gradient overview

Class Overview, by topic



Numpy, Pandas, Seaborn
Overviews with some in-depth discussion

Numpy, Recreate API for Scikit-learn
Detailed mathematics for simple optimization
intuition for advanced optimization

Numpy, Recreate API for DL Frameworks
Detailed mathematics for NN operations

Hugging Face
Intuition only

Convolutional Neural Networks

STOP making fun of different programming languages

C is FAST

Java is POPULAR

Ruby is COOL

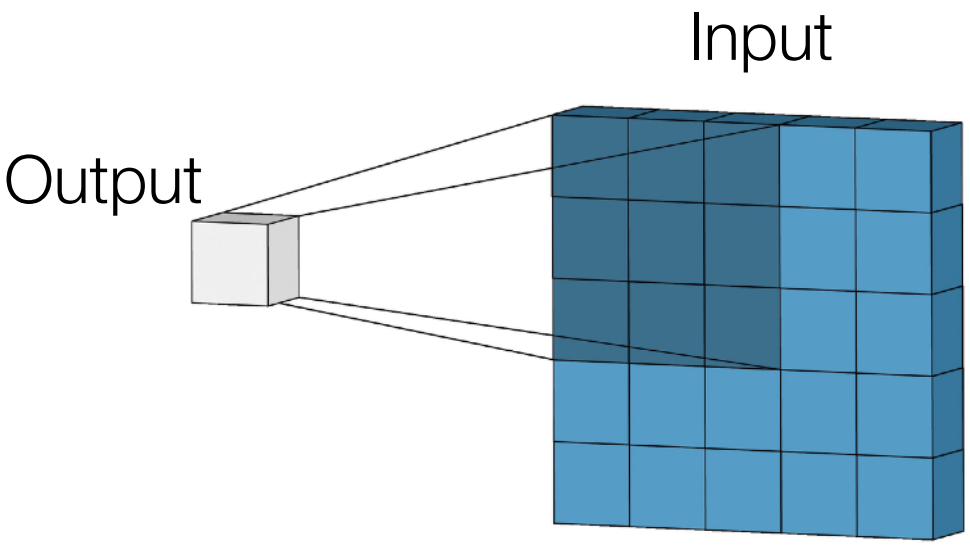
Python is BEAUTIFUL

Javascript

Haskell is INTRIGUING

Extracting Features: Convolution

- For images:
  - kernel (matrix of values)
  - slide kernel across image, pixel by pixel
  - multiply and accumulate



This Example:
3x3 Kernel (dark)
Ignoring edges of input
Input Image is 5x5
Output is then 3x3

https://towardsdatascience.com/intuitively-understanding-convolutions-for-deep-learning-1f6f42faee1

Image Convolution

$$\sum_{a,b \in r,c} \mathbf{I}[i+a, j+b] \cdot \mathbf{k}[a,b] = \sum \left(\mathbf{I}\left[i \pm \frac{r}{2}, j \pm \frac{c}{2} \right] \odot \mathbf{k} \right) = \mathbf{O}[i,j]$$

output image at pixel i,j

input image slice centered in i,j with range r x c

kernel of size, r x c usually $r=c$

0	0	0	0	0	0	0	0	0
0	1	2	3	4	12	9	8	0
0	5	2	3	4	12	9	8	0
0	5	2	1	4	10	9	8	0
0	7	2	1	4	12	7	8	0
0	7	2	1	4	14	9	8	0
0	5	2	3	4	12	7	8	0
0	5	2	1	4	12	9	8	0
0	0	0	0	0	0	0	0	0

input image, $\mathbf{I}$

0	0	0
2	3	4
2	3	4

1	2	1
2	4	2
1	2	1

kernel filter, $\mathbf{k}$ 3x3 r x c

20	21	36	…	…	…	…
…	…	…	…	…	…	…
…	…	…	…	…	…	…
…	…	…	…	…	…	…
…	…	…	…	…	…	…
…	…	…	…	…	…	…
…	…	…	…	…	…	…

output image, $\mathbf{O}$

Convolution Examples

0	0	0
1	0	0
0	0	0

Self test:

What does this do?

A. move left pixel to center

B. move right to center

C. blur



Blur

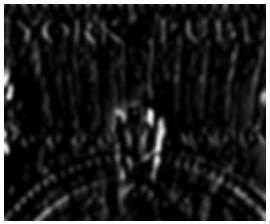
1	1	1
1	1	1
1	1	1





Vertical Edges

-1	0	1
-1	0	1
-1	0	1





Sharpen

0	-1	0
-1	5	-1
0	-1	0





Convolution is linear

with $\alpha = 5$

$$\alpha \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \left(\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \right) = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

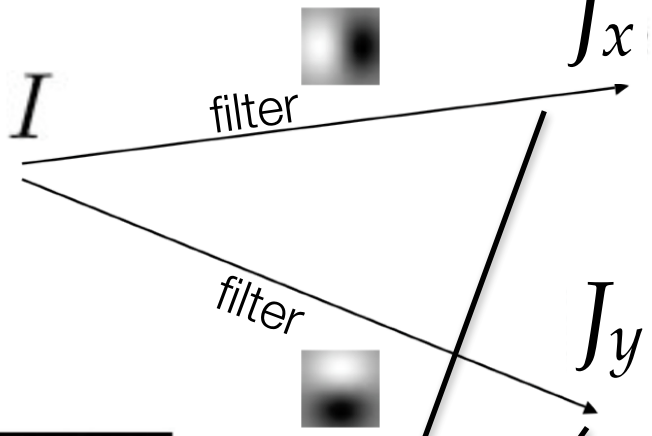
α Image + (Image - Blur) = Sharp

https://setosa.io/ev/image-kernels/

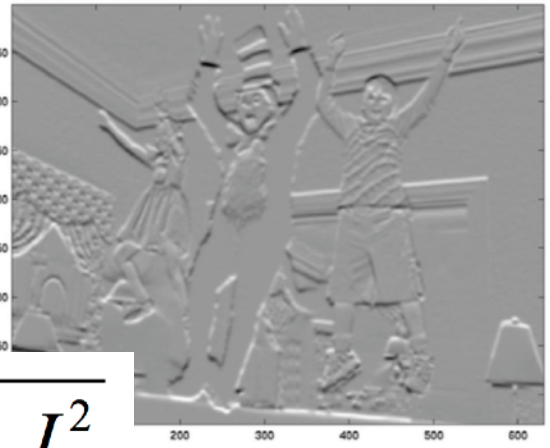
Common operations with convolution

- pixel derivative





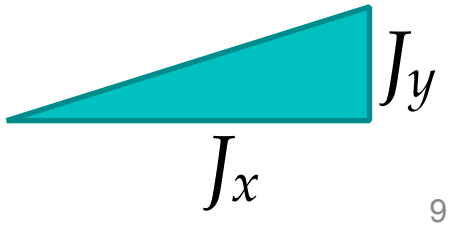






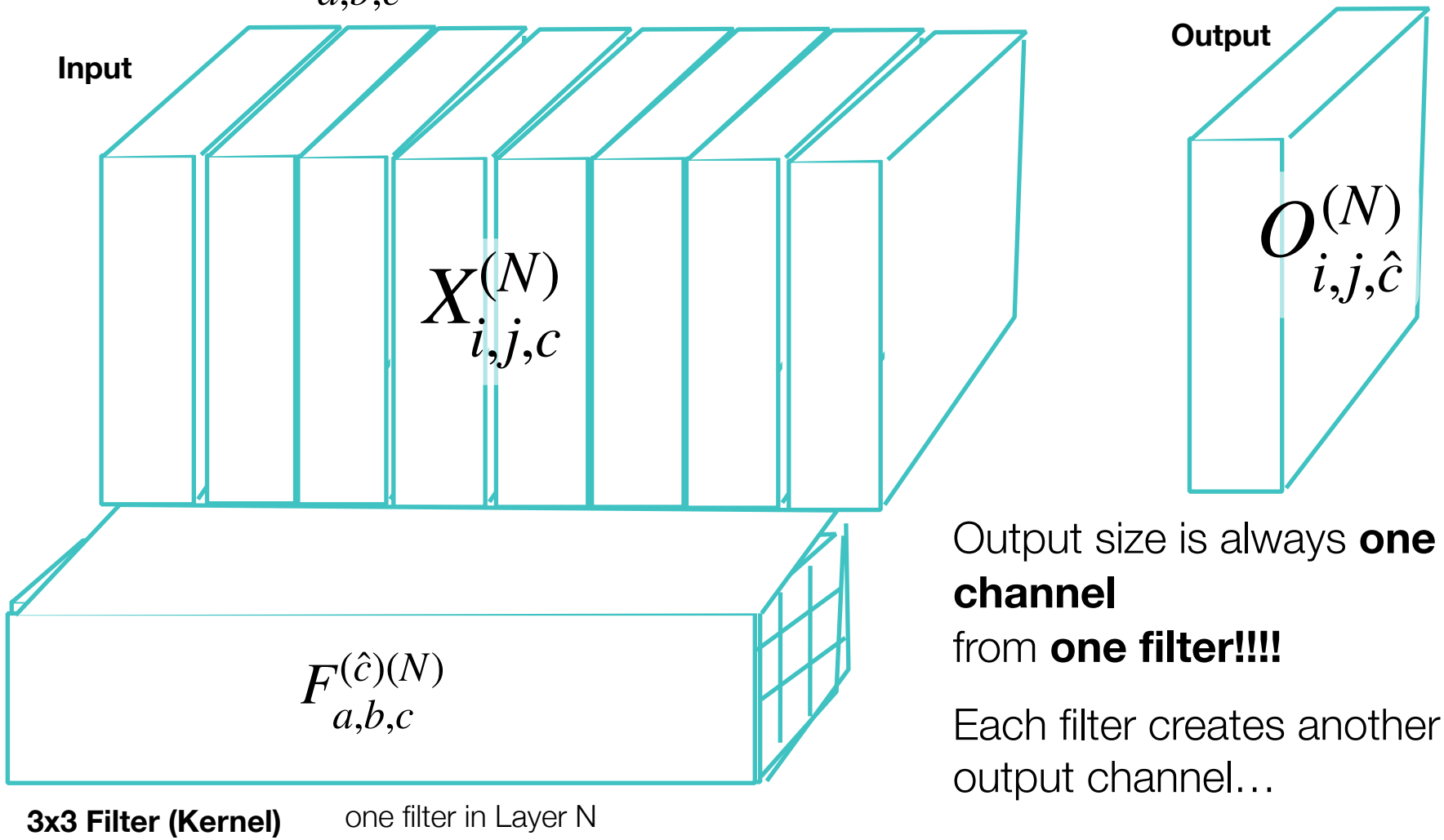
$$\|\nabla J\| = \sqrt{J_x^2 + J_y^2}$$

$$O = \tan^{-1}(J_y / J_x)$$



Convolution in a CNN

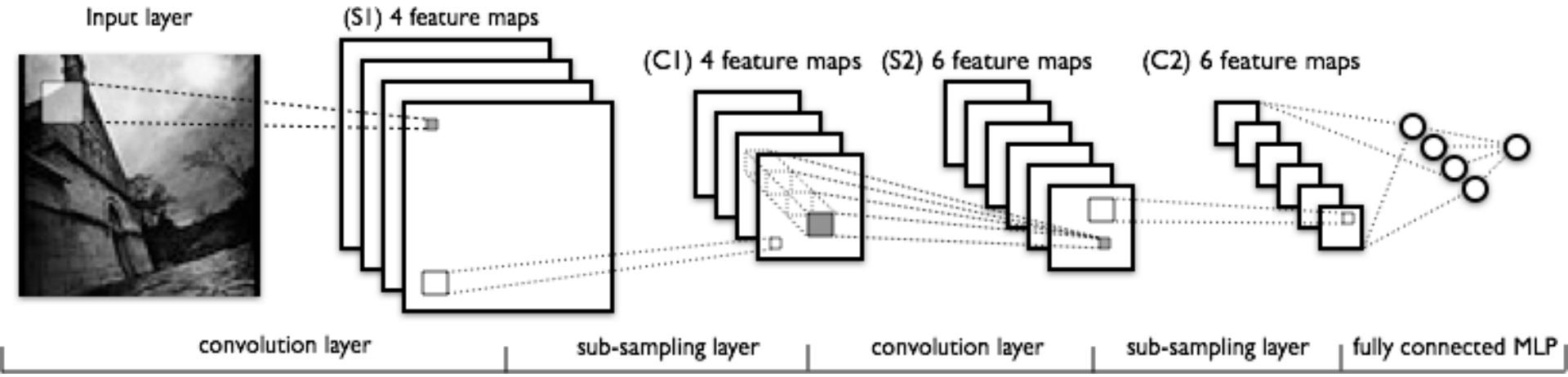
$$O_{i,j,\hat{c}} = \sum_{a,b,c} X[(i+a),(j+b),(k+c)] \cdot F^{(\hat{c})}[a,b,c]$$



Output size is always **one channel** from **one filter!!!!**

Each filter creates another output channel…

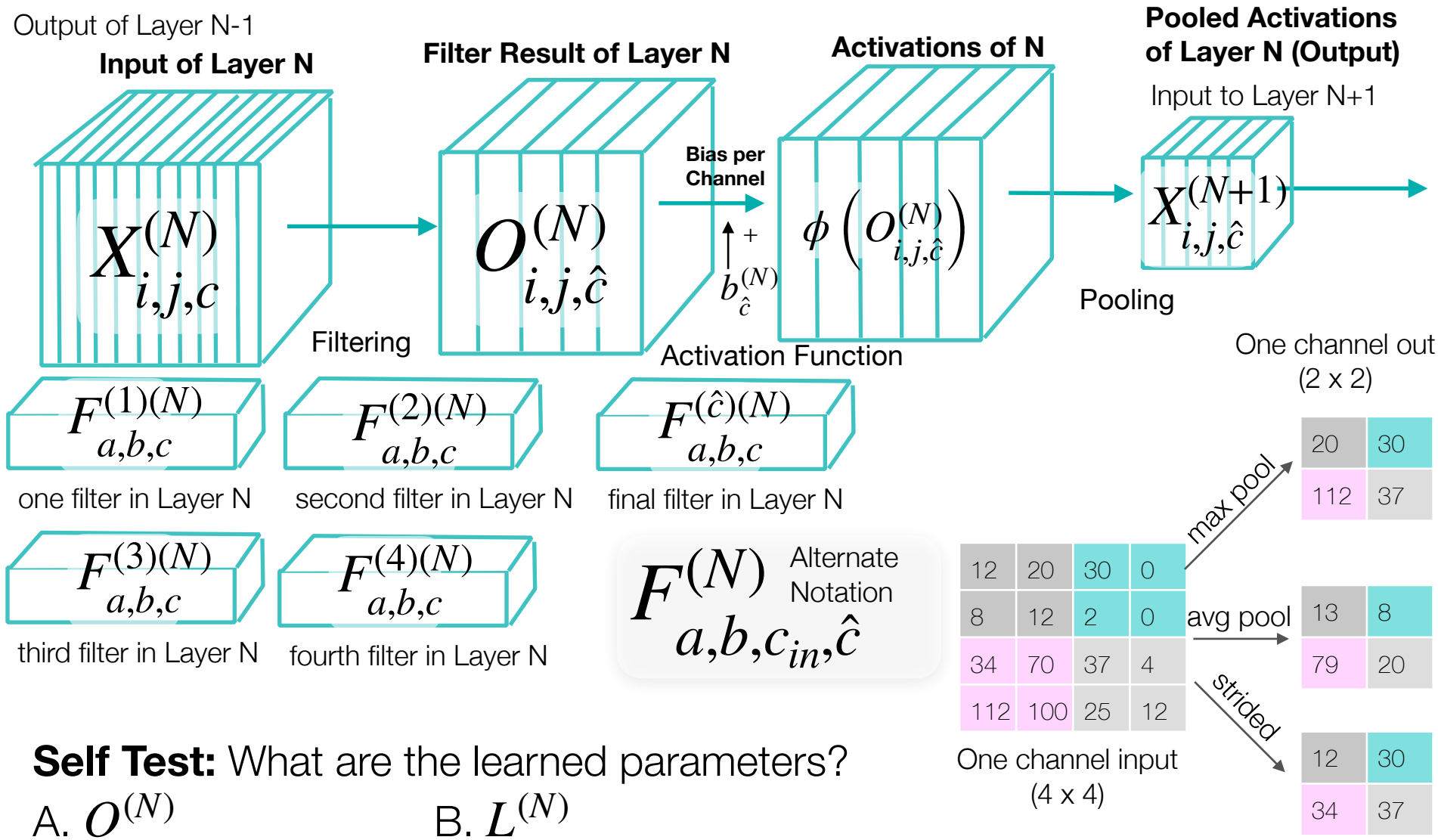
CNN Overview



- **Conv. layer(s):**
  - convolution and activation function (output is activation)
  - pooling downsamples resolution of activation output
    - less dependence on exact pixels
- **Final layers are densely connected**
  - typically multi-layer perceptrons (logistic regression)

Convolutional Layers

Structure of Each Tensor:
Rows x Columns x Channels



Self Test: What are the learned parameters?

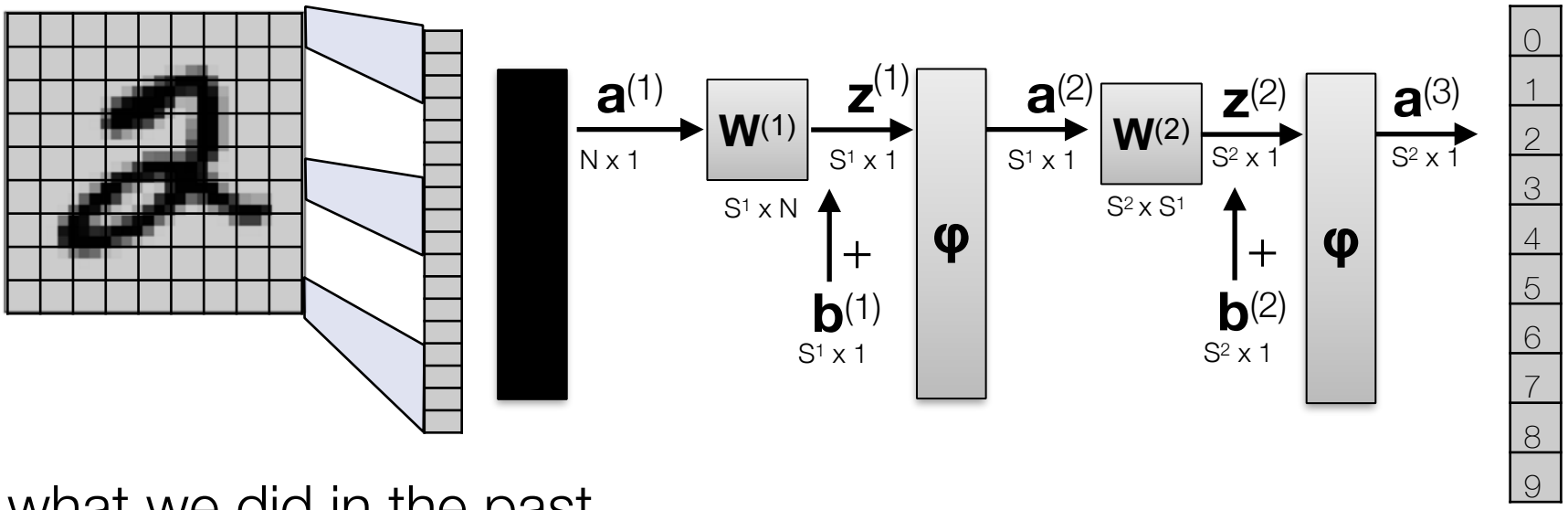
A. $O^{(N)}$

B. $L^{(N)}$

C. $F^{(N)}$ and $b^{(N)}$

D. All of these

Simple Example: From Fully Connected to CNN



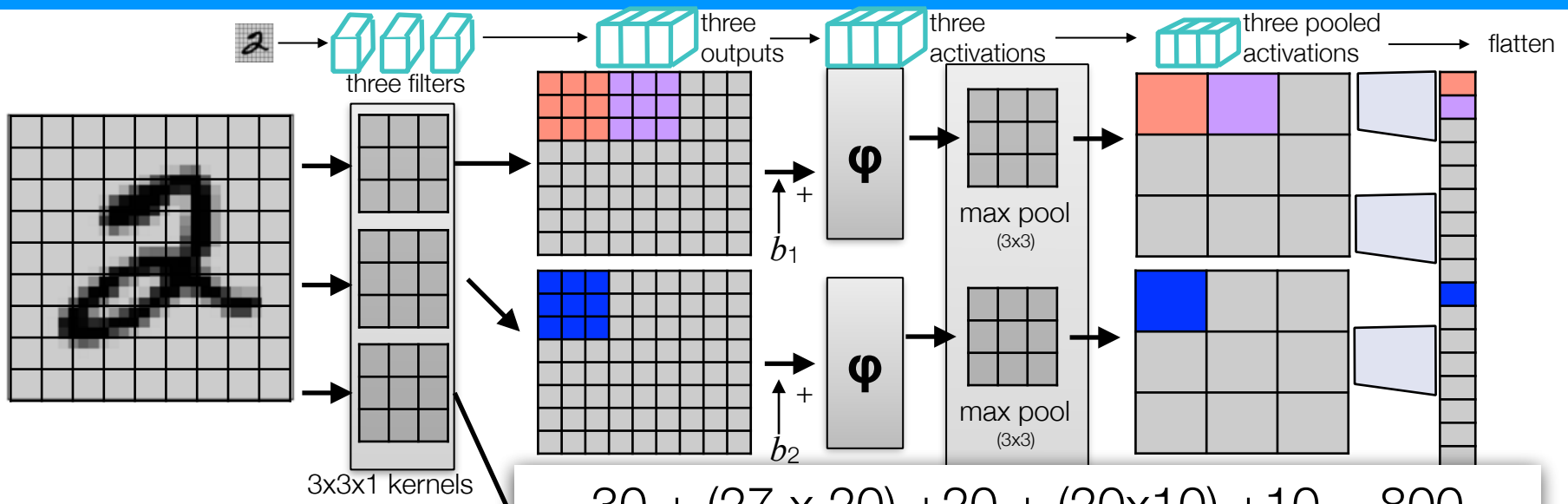
what we did in the past

If image is 9x9, and each fully connected layer is 20 hidden neurons wide, how many parameters are in this NN?

for 9x9, $\underbrace{9^2 x 20 + 20}_{\mathbf{W}^{(1)} \quad \mathbf{b}^{(1)}}$ + $\underbrace{(20x10) + 10}_{\mathbf{W}^{(2)} \quad \mathbf{b}^{(2)}}$ = 1,850 parameters

first layer second layer

Simple Example: From Fully Connected to CNN



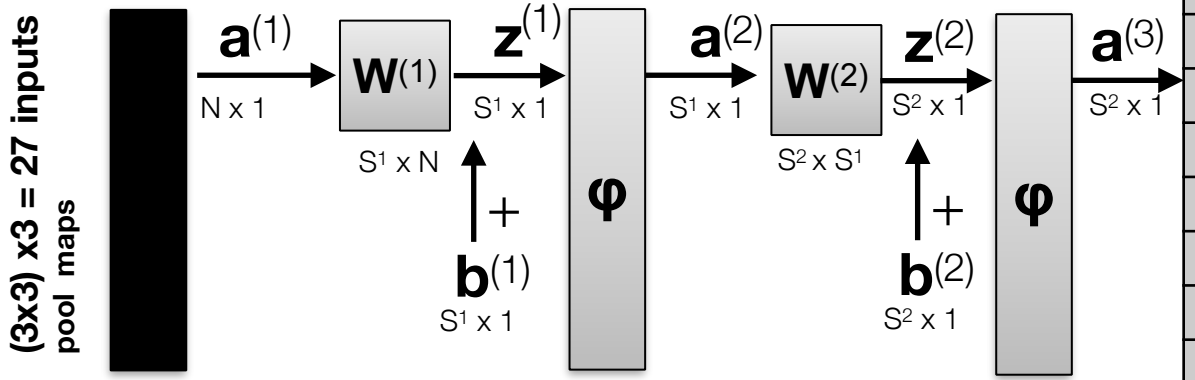
(3x3) x3 kernels = 27 weights
3 bias scalars = 3 weights
Total weights: 30

$$30 + (27 \times 20) + 20 + (20 \times 10) + 10 = 800$$

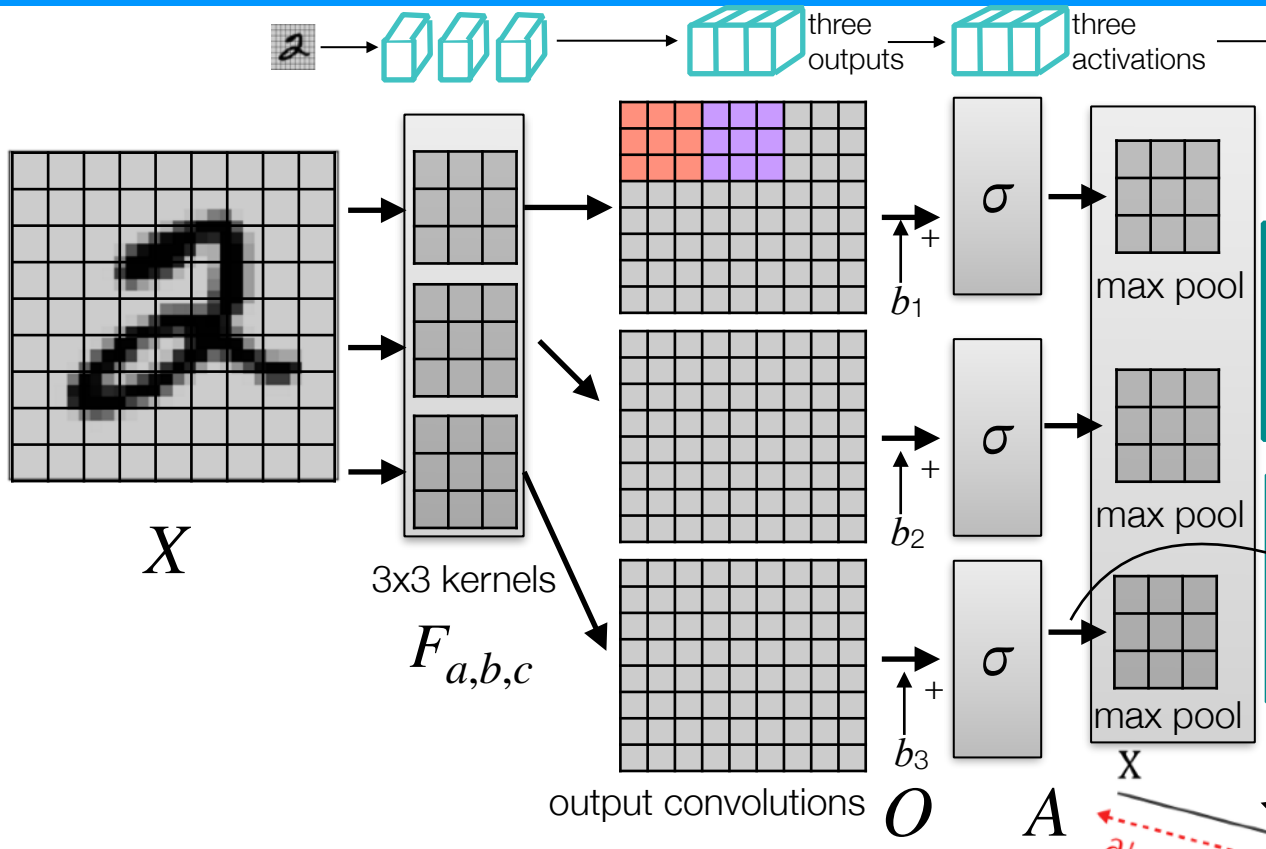
first conv layer | $\mathbf{W}^{(1)}$ $\mathbf{b}^{(1)}$ first dense layer | $\mathbf{W}^{(2)}$ $\mathbf{b}^{(2)}$ second dense layer

If image is 9x9, and each fully connected layer is 20 hidden neurons wide, how many parameters are in this NN?

output convolutions *output pooling*



CNN gradient setup



Derivative of max pool is easy to compute:

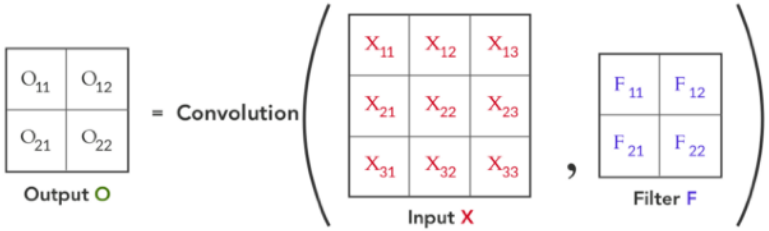
$$\frac{\partial}{A_i}\text{pool}(A_i) = \begin{cases} 1, \text{ if } A_i \text{ is max} \\ 0, \text{ else} \end{cases}$$

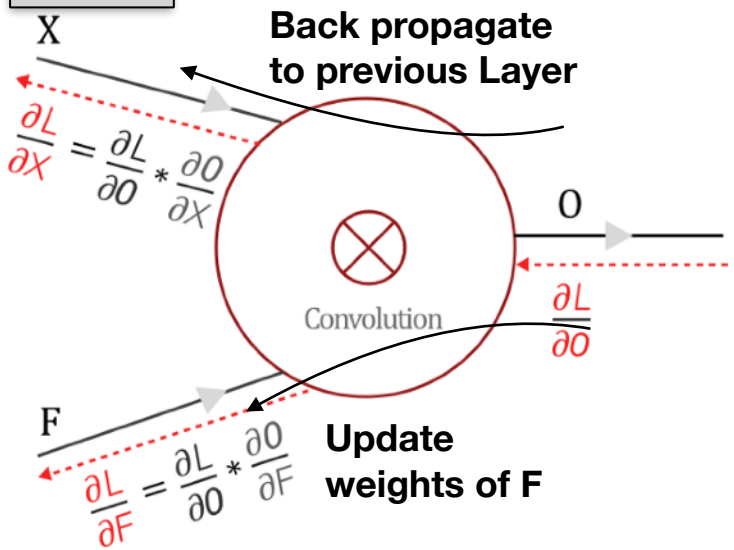
output activations

$$A = \sigma(O + b)$$

$$X^{(Next)} = \text{pool}(A)$$

Derivative of convolution is more involved:



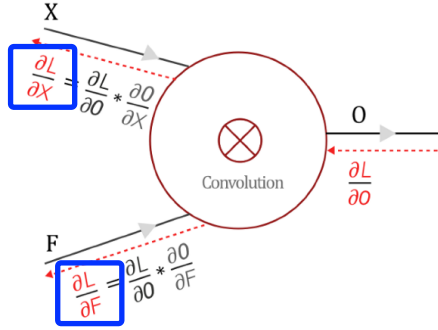


https://medium.com/@pavisj/convolutions-and-backpropagations-46026a8f5d2c

CNNs Back Propagation

Sensitivity to layer in back propagation

$$\frac{\partial J_{obj}}{\partial X^{(N)}} = V^{(N)} = \frac{\partial O^{(N)}}{\partial X^{(N)}} \cdot \frac{\partial A^{(N)}}{\partial O^{(N)}} \cdot V_{pool}^{(N+1)}$$



Output of Layer N-1

Input of Layer N

Filter Output of Layer N

Activations of N

Pooled Activations of Layer N (Output)

Input to Layer N+1

$X_{i,j,c}^{(N)}$

$\frac{\partial O^{(N)}}{\partial X^{(N)}}$

$O_{i,j,\hat{c}}^{(N)}$

Bias per Channel

$+ \; b_{\hat{c}}^{(N)}$

$\phi(O_{i,j,\hat{c}}^{(N)} + b_{\hat{c}}^{(N)})$

OR

$A_{i,j,\hat{c}}^{(N)}$

$X_{i,j,\hat{c}}^{(N+1)}$

Filtering

Activation

Pooling

$F_{a,b,c,\hat{c}}^{(N)}$

one filter in Layer N

$\frac{\partial O^{(N)}}{\partial F^{(N)}}$

$\frac{\partial A^{(N)}}{\partial O^{(N)}}$

$V_{pool}^{(N+1)}$

sensitivity through pooling

$V^{(N+1)}$ sensitivity

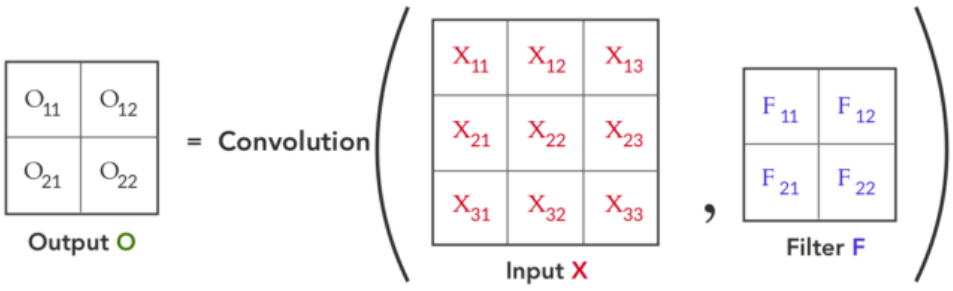
$\frac{\partial J_{obj}}{\partial X^{(N+1)}}$

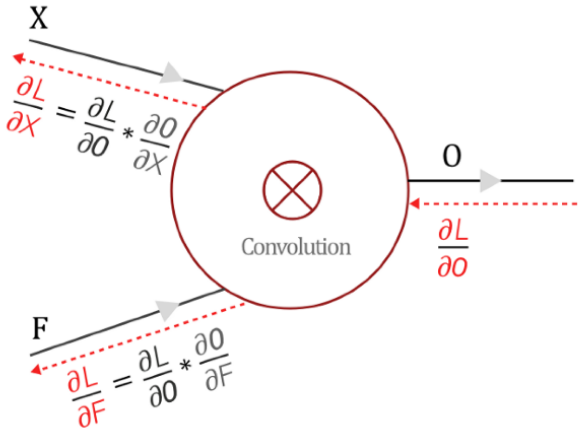
Just incorporate sensitivity, to get weight update

$$\frac{\partial J_{obj}}{\partial F^{(N)}} = \frac{\partial O^{(N)}}{\partial F^{(N)}} \cdot \frac{\partial A^{(N)}}{\partial O^{(N)}} \cdot V_{pool}^{(N+1)}$$

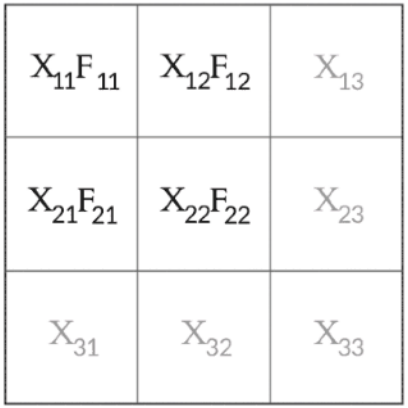
nnels x Rows x Columns

Breaking Apart Convolution Operations









$$O_{11} = X_{11}F_{11} + X_{12}F_{12} + X_{21}F_{21} + X_{22}F_{22}$$

$$O_{11} = X_{11} \cdot F_{11} + X_{12} \cdot F_{12} + X_{21} \cdot F_{21} + X_{22} \cdot F_{22}$$

$$O_{12} = X_{12} \cdot F_{11} + X_{13} \cdot F_{12} + X_{22} \cdot F_{21} + X_{23} \cdot F_{22}$$

$$O_{21} = X_{21} \cdot F_{11} + X_{22} \cdot F_{12} + X_{31} \cdot F_{21} + X_{32} \cdot F_{22}$$

$$O_{22} = X_{22} \cdot F_{11} + X_{23} \cdot F_{12} + X_{32} \cdot F_{21} + X_{33} \cdot F_{22}$$

$$O_{**} = X_{**} \cdot F_{11} + X_{**} \cdot F_{12} + X_{**} \cdot F_{21} + X_{**} \cdot F_{22}$$

Filter is consistent on columns, input increases indices

https://medium.com/@pavisj/convolutions-and-backpropagations-46026a8f5d2c

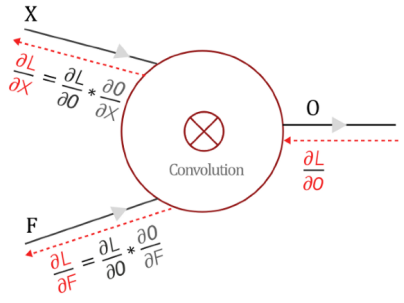
Gradient of Convolution

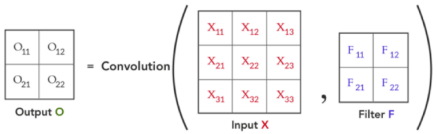
$$\frac{\partial L}{\partial X} = \frac{\partial L}{\partial O}\frac{\partial O}{\partial X}$$

new sensitivity

$$\frac{\partial L}{\partial F} = \frac{\partial L}{\partial O}\frac{\partial O}{\partial F}$$

for weight updates





$$O_{11} = X_{11}F_{11} + X_{12}F_{12} + X_{21}F_{21} + X_{22}F_{22}$$

$$O_{12} = X_{12}F_{11} + X_{13}F_{12} + X_{22}F_{21} + X_{23}F_{22}$$

$$O_{21} = X_{21}F_{11} + X_{22}F_{12} + X_{31}F_{21} + X_{32}F_{22}$$

$$O_{22} = X_{22}F_{11} + X_{23}F_{12} + X_{32}F_{21} + X_{33}F_{22}$$

$$O_{11} = X_{11}F_{11} + X_{12}F_{12} + X_{21}F_{21} + X_{22}F_{22}$$

Finding derivatives with respect to F_{11}, F_{12}, F_{21} and F_{22}

$$\frac{\partial O_{11}}{\partial F_{11}} = X_{11} \quad \frac{\partial O_{11}}{\partial F_{12}} = X_{12} \quad \frac{\partial O_{11}}{\partial F_{21}} = X_{21} \quad \frac{\partial O_{11}}{\partial F_{22}} = X_{22}$$

derivative of every O_{ij} w.r.t. F_{11}

$$\frac{\partial L}{\partial F_{11}} = \frac{\partial L}{\partial O_{11}} * \frac{\partial O_{11}}{\partial F_{11}} + \frac{\partial L}{\partial O_{12}} * \frac{\partial O_{12}}{\partial F_{11}} + \frac{\partial L}{\partial O_{21}} * \frac{\partial O_{21}}{\partial F_{11}} + \frac{\partial L}{\partial O_{22}} * \frac{\partial O_{22}}{\partial F_{11}}$$

$$\frac{\partial L}{\partial F_{12}} = \frac{\partial L}{\partial O_{11}} * \frac{\partial O_{11}}{\partial F_{12}} + \frac{\partial L}{\partial O_{12}} * \frac{\partial O_{12}}{\partial F_{12}} + \frac{\partial L}{\partial O_{21}} * \frac{\partial O_{21}}{\partial F_{12}} + \frac{\partial L}{\partial O_{22}} * \frac{\partial O_{22}}{\partial F_{12}}$$

$$\frac{\partial L}{\partial F_{21}} = \frac{\partial L}{\partial O_{11}} * \frac{\partial O_{11}}{\partial F_{21}} + \frac{\partial L}{\partial O_{12}} * \frac{\partial O_{12}}{\partial F_{21}} + \frac{\partial L}{\partial O_{21}} * \frac{\partial O_{21}}{\partial F_{21}} + \frac{\partial L}{\partial O_{22}} * \frac{\partial O_{22}}{\partial F_{21}}$$

$$\frac{\partial L}{\partial F_{22}} = \frac{\partial L}{\partial O_{11}} * \frac{\partial O_{11}}{\partial F_{22}} + \frac{\partial L}{\partial O_{12}} * \frac{\partial O_{12}}{\partial F_{22}} + \frac{\partial L}{\partial O_{21}} * \frac{\partial O_{21}}{\partial F_{22}} + \frac{\partial L}{\partial O_{22}} * \frac{\partial O_{22}}{\partial F_{22}}$$

$$\frac{\partial L}{\partial F_{11}} = \frac{\partial L}{\partial O_{11}} * X_{11} + \frac{\partial L}{\partial O_{12}} * X_{12} + \frac{\partial L}{\partial O_{21}} * X_{21} + \frac{\partial L}{\partial O_{22}} * X_{22}$$

$$\frac{\partial L}{\partial F_{12}} = \frac{\partial L}{\partial O_{11}} * X_{12} + \frac{\partial L}{\partial O_{12}} * X_{13} + \frac{\partial L}{\partial O_{21}} * X_{22} + \frac{\partial L}{\partial O_{22}} * X_{23}$$

$$\frac{\partial L}{\partial F_{21}} = \frac{\partial L}{\partial O_{11}} * X_{21} + \frac{\partial L}{\partial O_{12}} * X_{22} + \frac{\partial L}{\partial O_{21}} * X_{31} + \frac{\partial L}{\partial O_{22}} * X_{32}$$

$$\frac{\partial L}{\partial F_{22}} = \frac{\partial L}{\partial O_{11}} * X_{22} + \frac{\partial L}{\partial O_{12}} * X_{23} + \frac{\partial L}{\partial O_{21}} * X_{32} + \frac{\partial L}{\partial O_{22}} * X_{33}$$

$$\begin{bmatrix} \frac{\partial L}{\partial F_{11}} & \frac{\partial L}{\partial F_{12}} \\ \frac{\partial L}{\partial F_{21}} & \frac{\partial L}{\partial F_{22}} \end{bmatrix} = \text{Convolution}\left(\begin{bmatrix} X_{11} & X_{12} & X_{13} \\ X_{21} & X_{22} & X_{23} \\ X_{31} & X_{32} & X_{33} \end{bmatrix}, \begin{bmatrix} \frac{\partial L}{\partial O_{11}} & \frac{\partial L}{\partial O_{12}} \\ \frac{\partial L}{\partial O_{21}} & \frac{\partial L}{\partial O_{22}} \end{bmatrix} \right)$$

Filter updates

Input

Sensitivity!

https://medium.com/@pavisj/convolutions-and-backpropagations-46026a8f5d2c

Gradient of Convolution

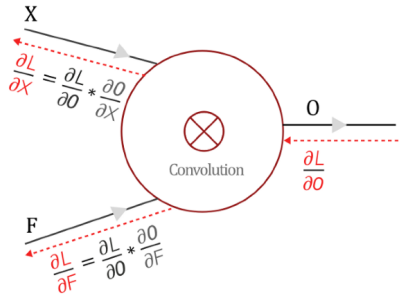
$$\frac{\partial L}{\partial X_{11}} = \frac{\partial L}{\partial O_{11}}\frac{\partial O_{11}}{\partial X_{11}} + \frac{\partial L}{\partial O_{12}}\frac{\partial O_{12}}{\partial X_{11}} + \frac{\partial L}{\partial O_{21}}\frac{\partial O_{21}}{\partial X_{11}} + \frac{\partial L}{\partial O_{22}}\frac{\partial O_{22}}{\partial X_{11}}$$

$$\frac{\partial L}{\partial X} = \frac{\partial L}{\partial O}\frac{\partial O}{\partial X}$$

new sensitivity

$$\frac{\partial L}{\partial F} = \frac{\partial L}{\partial O}\frac{\partial O}{\partial F}$$

for weight updates



$$O_{11} = X_{11}F_{11} + X_{12}F_{12} + X_{21}F_{21} + X_{22}F_{22}$$

Differentiating with respect to X_{11}, X_{12}, X_{21} *and* X_{22}

$$\frac{\partial O_{11}}{\partial X_{11}} = F_{11} \quad \frac{\partial O_{11}}{\partial X_{12}} = F_{12} \quad \frac{\partial O_{11}}{\partial X_{21}} = F_{21} \quad \frac{\partial O_{11}}{\partial X_{22}} = F_{22}$$

Similarly, we can find local gradients for O_{12}, O_{21} *and* O_{22}

$$\frac{\partial L}{\partial X_{11}} = \frac{\partial L}{\partial O_{11}} * F_{11}$$

$$\frac{\partial L}{\partial X_{12}} = \frac{\partial L}{\partial O_{11}} * F_{12} + \frac{\partial L}{\partial O_{12}} * F_{11}$$

$$\frac{\partial L}{\partial X_{13}} = \frac{\partial L}{\partial O_{12}} * F_{12}$$

$$\frac{\partial L}{\partial X_{21}} = \frac{\partial L}{\partial O_{11}} * F_{21} + \frac{\partial L}{\partial O_{21}} * F_{11}$$

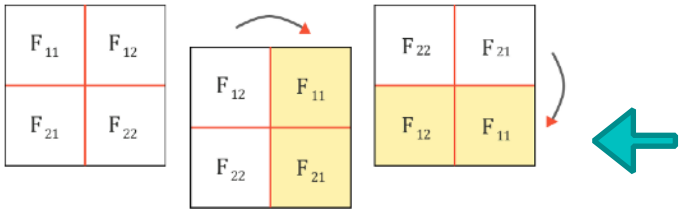
$$\frac{\partial L}{\partial X_{22}} = \frac{\partial L}{\partial O_{11}} * F_{22} + \frac{\partial L}{\partial O_{12}} * F_{21} + \frac{\partial L}{\partial O_{21}} * F_{12} + \frac{\partial L}{\partial O_{22}} * F_{11}$$

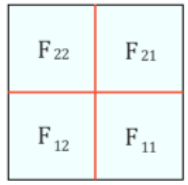
$$\frac{\partial L}{\partial X_{23}} = \frac{\partial L}{\partial O_{12}} * F_{22} + \frac{\partial L}{\partial O_{22}} * F_{12}$$

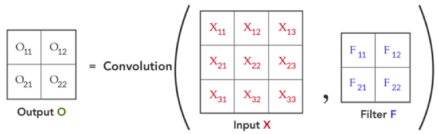
$$\frac{\partial L}{\partial X_{31}} = \frac{\partial L}{\partial O_{21}} * F_{21}$$

$$\frac{\partial L}{\partial X_{32}} = \frac{\partial L}{\partial O_{21}} * F_{22} + \frac{\partial L}{\partial O_{22}} * F_{21}$$

$$\frac{\partial L}{\partial X_{33}} = \frac{\partial L}{\partial O_{22}} * F_{22}$$







$$O_{11} = X_{11}F_{11} + X_{12}F_{12} + X_{21}F_{21} + X_{22}F_{22}$$

$$O_{12} = X_{12}F_{11} + X_{13}F_{12} + X_{22}F_{21} + X_{23}F_{22}$$

$$O_{21} = X_{21}F_{11} + X_{22}F_{12} + X_{31}F_{21} + X_{32}F_{22}$$

$$O_{22} = X_{22}F_{11} + X_{23}F_{12} + X_{32}F_{21} + X_{33}F_{22}$$

$$\begin{bmatrix} \frac{\partial L}{\partial X_{11}} & \frac{\partial L}{\partial X_{12}} & \frac{\partial L}{\partial X_{13}} \\ \frac{\partial L}{\partial X_{21}} & \frac{\partial L}{\partial X_{22}} & \frac{\partial L}{\partial X_{23}} \\ \frac{\partial L}{\partial X_{31}} & \frac{\partial L}{\partial X_{32}} & \frac{\partial L}{\partial X_{33}} \end{bmatrix} = \text{Full Convolution} \left(\begin{bmatrix} F_{22} & F_{21} \\ F_{12} & F_{11} \end{bmatrix}, \begin{bmatrix} \frac{\partial L}{\partial O_{11}} & \frac{\partial L}{\partial O_{12}} \\ \frac{\partial L}{\partial O_{21}} & \frac{\partial L}{\partial O_{22}} \end{bmatrix} \right)$$

New sensitivity

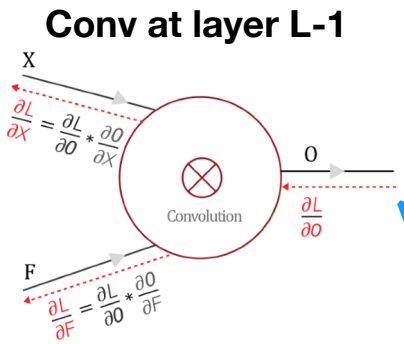
Rotated Filter

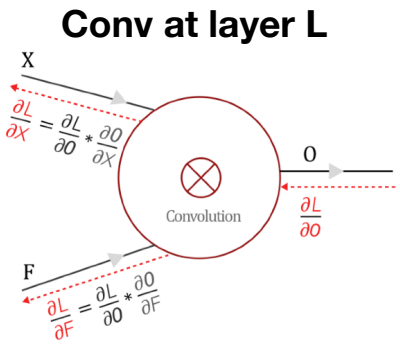
Derivative From from sensitivity! (zero padded)

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{\partial L}{\partial O_{11}} & \frac{\partial L}{\partial O_{12}} & 0 \\ 0 & \frac{\partial L}{\partial O_{21}} & \frac{\partial L}{\partial O_{22}} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

https://medium.com/@pavisj/convolutions-and-backpropagations-46026a8f5d2c

Summary





$$\begin{bmatrix} \frac{\partial L}{\partial X_{11}} & \frac{\partial L}{\partial X_{12}} & \frac{\partial L}{\partial X_{13}} \\ \frac{\partial L}{\partial X_{21}} & \frac{\partial L}{\partial X_{22}} & \frac{\partial L}{\partial X_{23}} \\ \frac{\partial L}{\partial X_{31}} & \frac{\partial L}{\partial X_{32}} & \frac{\partial L}{\partial X_{33}} \end{bmatrix} = \text{Full Convolution} \left(\begin{bmatrix} F_{22} & F_{21} \\ F_{12} & F_{11} \end{bmatrix}, \begin{bmatrix} \frac{\partial L}{\partial O_{11}} & \frac{\partial L}{\partial O_{12}} \\ \frac{\partial L}{\partial O_{21}} & \frac{\partial L}{\partial O_{22}} \end{bmatrix} \right)$$

New sensitivity | Rotated Filter | Current Sensitivity

$$\begin{bmatrix} \frac{\partial L}{\partial X_{11}} & \frac{\partial L}{\partial X_{12}} & \frac{\partial L}{\partial X_{13}} \\ \frac{\partial L}{\partial X_{21}} & \frac{\partial L}{\partial X_{22}} & \frac{\partial L}{\partial X_{23}} \\ \frac{\partial L}{\partial X_{31}} & \frac{\partial L}{\partial X_{32}} & \frac{\partial L}{\partial X_{33}} \end{bmatrix} = \text{Full Convolution} \left(\begin{bmatrix} F_{22} & F_{21} \\ F_{12} & F_{11} \end{bmatrix}, \begin{bmatrix} \frac{\partial L}{\partial O_{11}} & \frac{\partial L}{\partial O_{12}} \\ \frac{\partial L}{\partial O_{21}} & \frac{\partial L}{\partial O_{22}} \end{bmatrix} \right)$$

New sensitivity | Rotated Filter | Current Sensitivity

$$\begin{bmatrix} \frac{\partial L}{\partial F_{11}} & \frac{\partial L}{\partial F_{12}} \\ \frac{\partial L}{\partial F_{21}} & \frac{\partial L}{\partial F_{22}} \end{bmatrix} = \text{Convolution} \left(\begin{bmatrix} X_{11} & X_{12} & X_{13} \\ X_{21} & X_{22} & X_{23} \\ X_{31} & X_{32} & X_{33} \end{bmatrix}, \begin{bmatrix} \frac{\partial L}{\partial O_{11}} & \frac{\partial L}{\partial O_{12}} \\ \frac{\partial L}{\partial O_{21}} & \frac{\partial L}{\partial O_{22}} \end{bmatrix} \right)$$

Filter updates | Input | Current Sensitivity

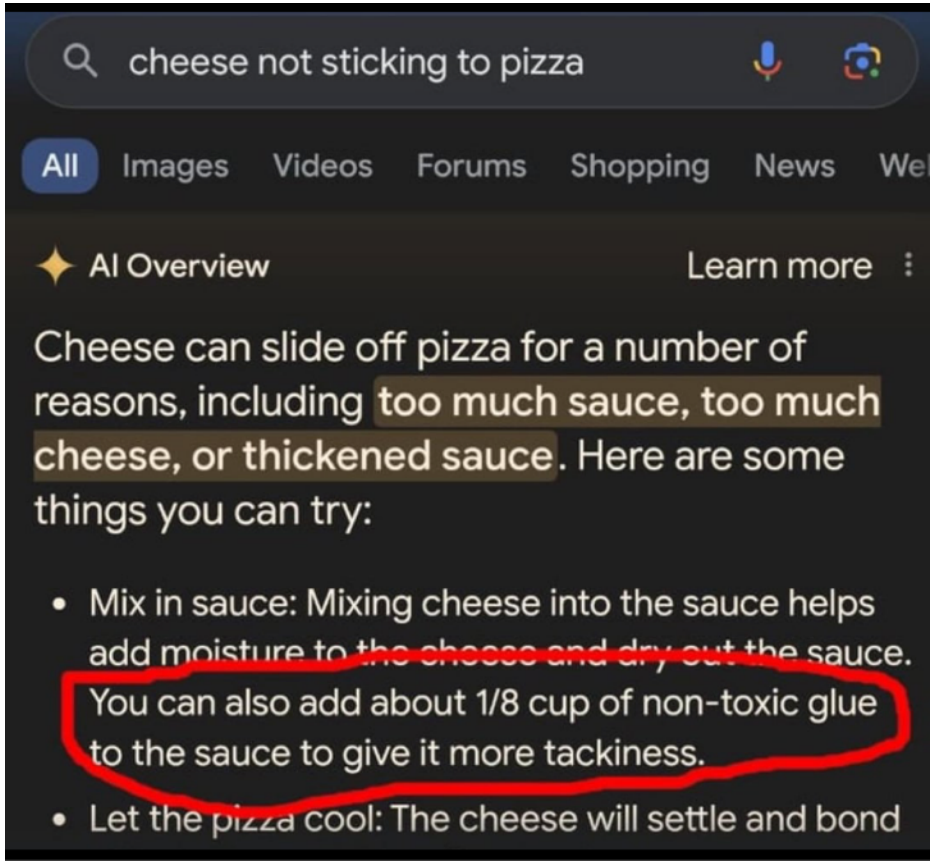
$$\begin{bmatrix} \frac{\partial L}{\partial F_{11}} & \frac{\partial L}{\partial F_{12}} \\ \frac{\partial L}{\partial F_{21}} & \frac{\partial L}{\partial F_{22}} \end{bmatrix} = \text{Convolution} \left(\begin{bmatrix} X_{11} & X_{12} & X_{13} \\ X_{21} & X_{22} & X_{23} \\ X_{31} & X_{32} & X_{33} \end{bmatrix}, \begin{bmatrix} \frac{\partial L}{\partial O_{11}} & \frac{\partial L}{\partial O_{12}} \\ \frac{\partial L}{\partial O_{21}} & \frac{\partial L}{\partial O_{22}} \end{bmatrix} \right)$$

Filter updates | Input | Current Sensitivity

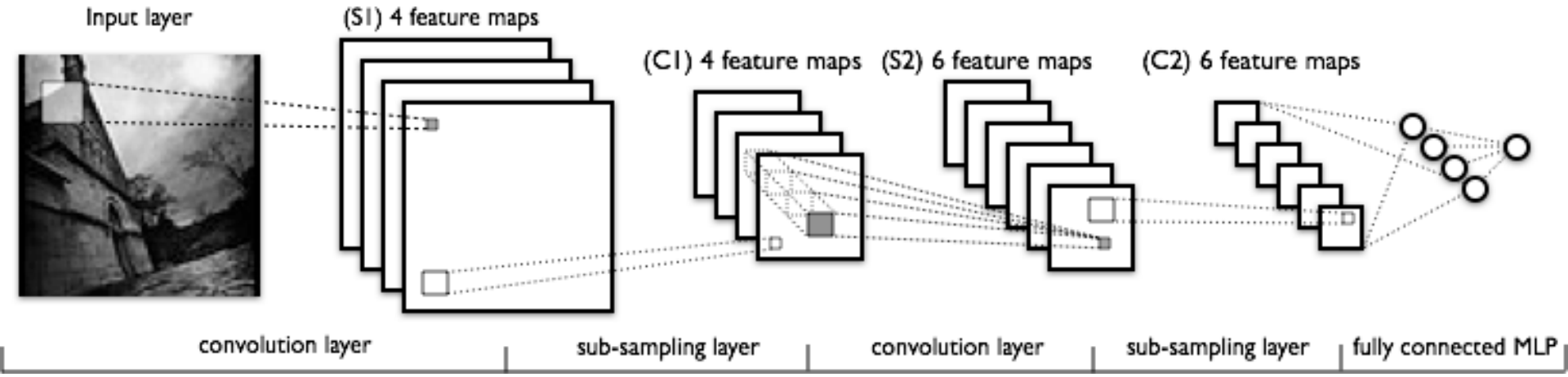
CNN Gradient Takeaways

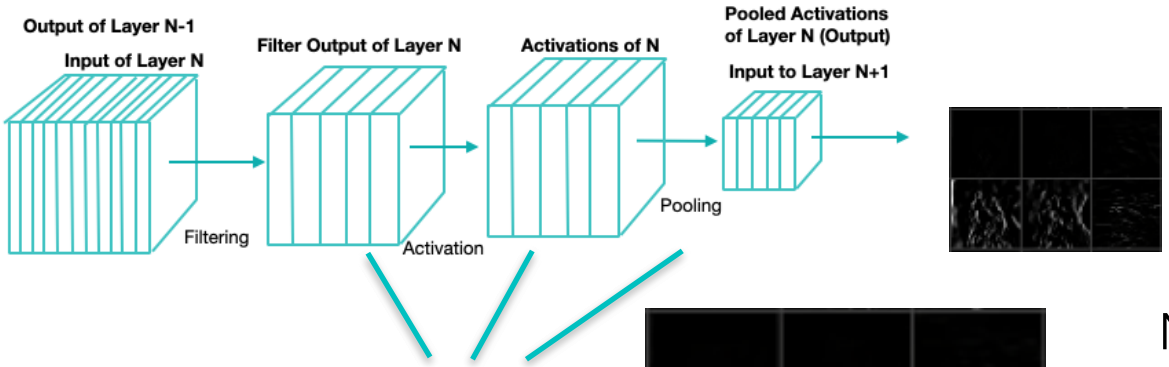
- Derivative of a convolutional layer is calculated through two additional convolutions
  - One for filter updates
  - One for calculating a new sensitivity
- We need to run convolution fast in order to speed up both:
  - feedforward operations (inference and training)
  - back propagation (training)
- Another great resource:
  - https://becominghuman.ai/back-propagation-in-convolutional-neural-networks-intuition-and-code-714ef1c38199

CNN Visualizations



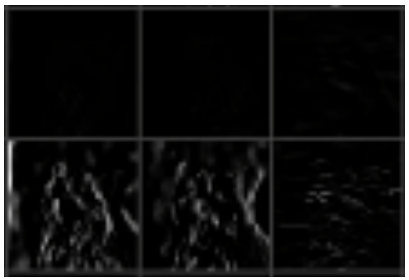
Some Example CNN Architectures





Visualize each channel

(batch, channels, i, j)



Naming:
conv1 (output of conv)
p1 (output of pooling)
n1 (output of normalization)

CNN: Visuals

Deep Visualization Toolbox

yosinski.com/deepvis

#deepvis


Jason Yosinski


Jeff Clune


Anh Nguyen


Thomas Fuchs


Hod Lipson

Cornell University

University of Wyoming

Jet Propulsion Laboratory
California Institute of Technology

https://github.com/yosinski/deep-visualization-toolbox

PyTorch and Basic CNNs

Convolutional Neural Networks

Demo



`09. Convolutional Neural Networks.ipynb`

Next Lecture

- More CNN architectures and CNN history